

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q11-0083

Monroe 100 kV  
July 01, 2013

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0083. The Interconnection Customer proposes a 112.2 MW solar photovoltaic facility interconnecting at the Monroe 100 kV substation in the Michigan area. The request was received on November 06, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 66 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,317,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Laclede to Weiler 100 KV  
Contingency: Fermi 500 kV/230 kV TX  
Loading: 106.0% (Rating: 174 MVA)  
DFAX: 23.2%  
Pre-project loading: 91.0%
2. Facility: Doyle Energy Facility to Toll Road 100 KV  
Contingency: Fermi 500/230kV XFMR  
Loading: 103.5% (Rating: 138 MVA)  
DFAX: 8.8%  
Pre-project loading: 96.3%
3. Facility: Bonita Springs to Toll Rd 100 KV Line  
Contingency: Fermi 500 kV/230 kV TX  
Loading: 100.5% (Rating: 142 MVA)  
DFAX: 8.8%  
Pre-project loading: 93.6%

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

### **4. Required Network Upgrades and Cost Estimates**

| Facility                                  | Required Upgrade           | Cost Estimate | Allocation to Project |
|-------------------------------------------|----------------------------|---------------|-----------------------|
| Laclede to Weiler 100 KV Line             | Reconductor line           | \$1,064,000   | 100% (\$1,064,000)    |
| Doyle Energy Facility to Toll Road 100 KV | Reconductor line           | \$1,691,000   | 100% (\$1,691,000)    |
| Bonita Springs to Toll Rd 100 KV Line     | Reconductor line           | \$2,562,000   | 100% (\$2,562,000)    |
|                                           | Total allocated to project |               | \$5,317,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.